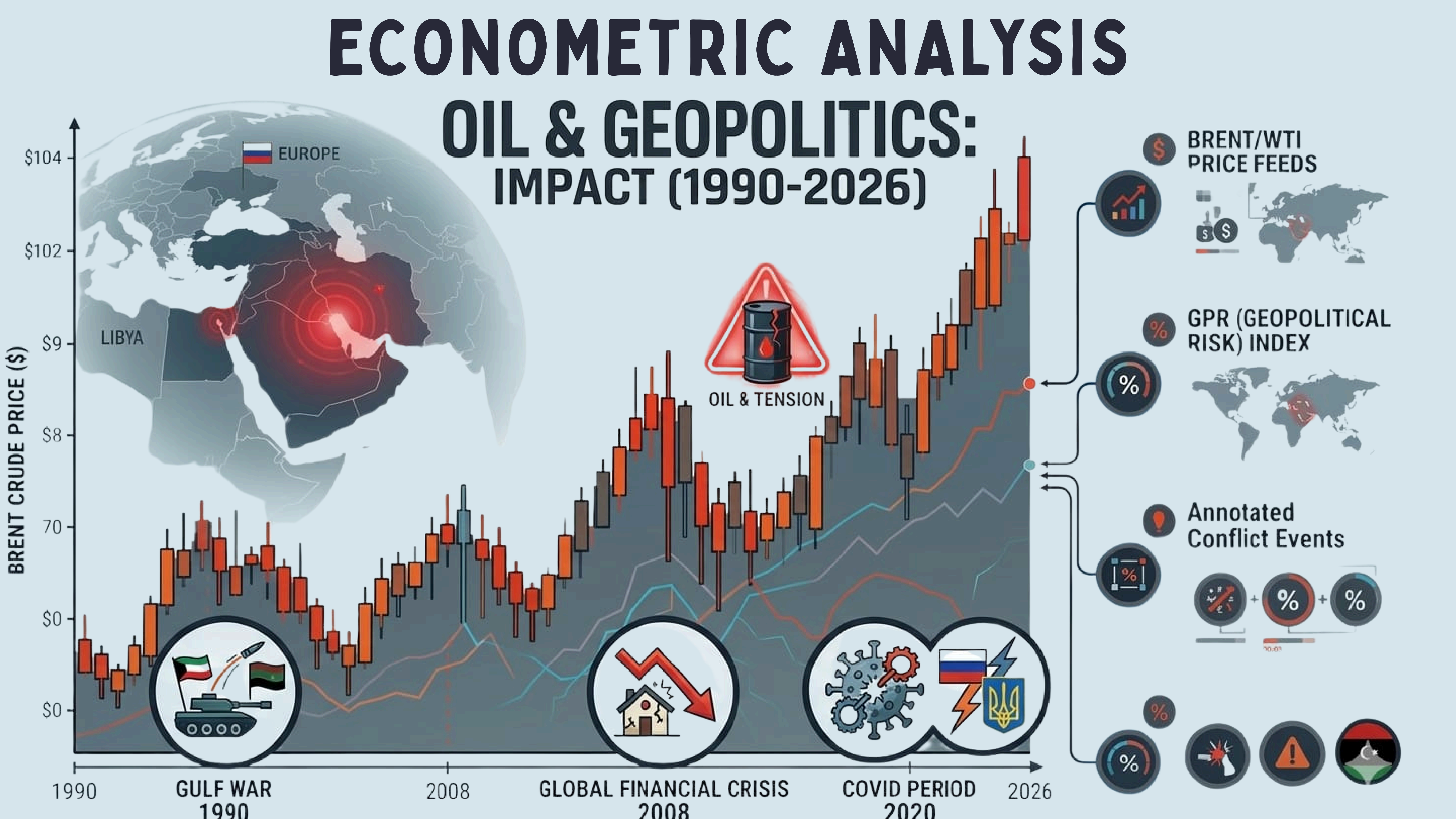

ECONOMETRIC ANALYSIS
OIL & GEOPOLITICS:
IMPACT (1990-2026)
EUROPE
LIBYA
OIL & TENSION
$104
$102
$9
$8
70
$0
$0
$0
BRENT CRUDE PRICE ($)
1990
2008
2026
GULF WAR
1990
GLOBAL FINANCIAL CRISIS
2008
COVID PERIOD
2020
BRENT/WTI
PRICE FEEDS
GPR (GEOPOLITICAL
RISK) INDEX
Annotated
Conflict Events

EVENT TIMELINE

1990 - 1991

GULF WAR

1

Our entrepreneurial story begins with a creative idea from three tech entrepreneurs.

2007 - 2008

GLOBAL FINANCIAL CRISIS

2

Stage where we focus on a process of analysis, study of the market and the target audience.

2019 - 2021

COVID 19

3

Team building and training phase, brainstorming, company objectives, and roadmap.

Ongoing

RUSSIA - UKRIANE WAR

4

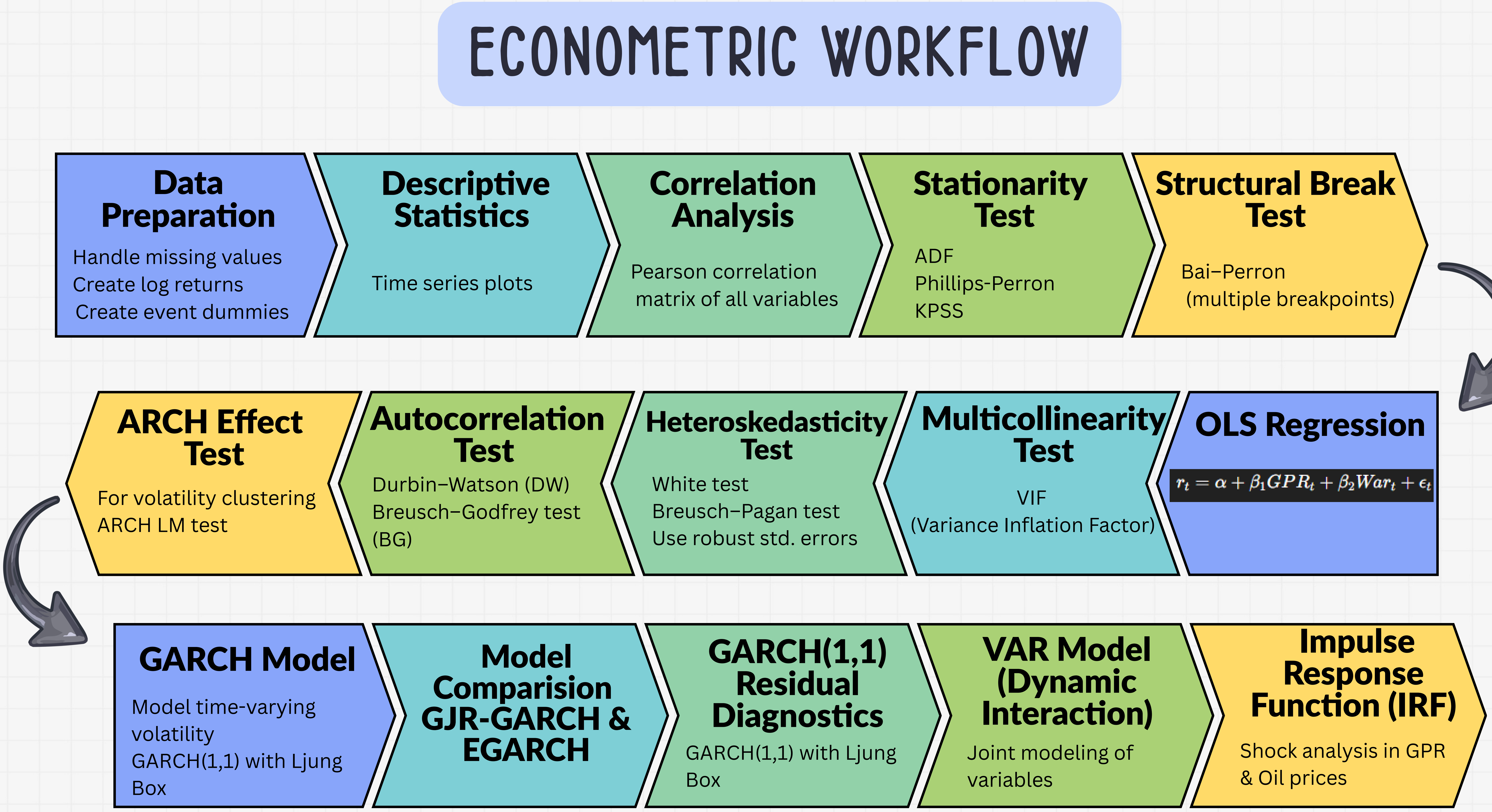
Launch of the first local technology project dedicated to universities and educational centers.

Ongoing

IRAN - ISRAEL USA WAR

5

Development of a digital solution focused on enhancing creativity in children.


ECONOMETRIC WORKFLOW
Data Preparation
Handle missing values
Create log returns
Create event dummies
Descriptive Statistics
Time series plots
Correlation Analysis
Pearson correlation matrix of all variables
Stationarity Test
ADF
Phillips-Perron
KPSS
Structural Break Test
Bai–Perron (multiple breakpoints)
OLS Regression
$r_t = \alpha + \beta_1 GPR_t + \beta_2 War_t + \epsilon_t$
Multicollinearity Test
VIF (Variance Inflation Factor)
Heteroskedasticity Test
White test
Breusch–Pagan test
Use robust std. errors
Autocorrelation Test
Durbin–Watson (DW)
Breusch–Godfrey test (BG)
ARCH Effect Test
For volatility clustering
ARCH LM test
GARCH Model
Model time-varying volatility
GARCH(1,1) with Ljung Box
Model Comparision GJR-GARCH & EGARCH
GARCH(1,1) Residual Diagnostics
GARCH(1,1) with Ljung Box
VAR Model (Dynamic Interaction)
Joint modeling of variables
Impulse Response Function (IRF)
Shock analysis in GPR & Oil prices

DATA INFO & EVENT DUMMIES

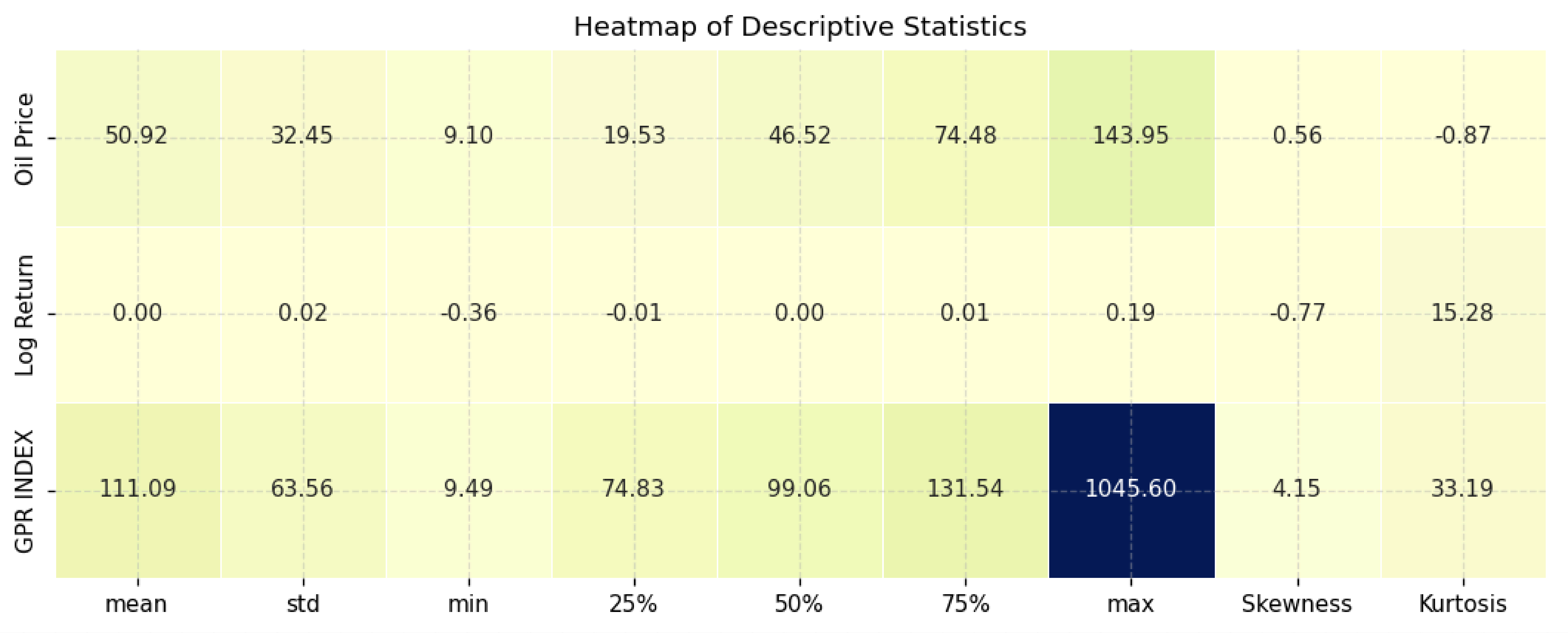
```
Dataset shape: (9828, 8)

Date range: 1987-05-20 to 2026-03-12

Event Counts:
Gulf_War               119
Oil_Crisis             144
Covid_Period           505
Russia_Ukraine_War    1019

First 5 rows:
```

	Date	Oil Price	Log Return	GPR INDEX	Gulf_War	Oil_Crisis	Covid_Period	Russia_Ukraine_War
0	1987-05-20	18.63	NaN	106.86	0	0	0	0
1	1987-05-21	18.45	-0.009709	77.52	0	0	0	0
2	1987-05-22	18.55	0.005405	132.50	0	0	0	0
3	1987-05-25	18.60	0.002692	87.69	0	0	0	0
4	1987-05-26	18.63	0.001612	82.44	0	0	0	0

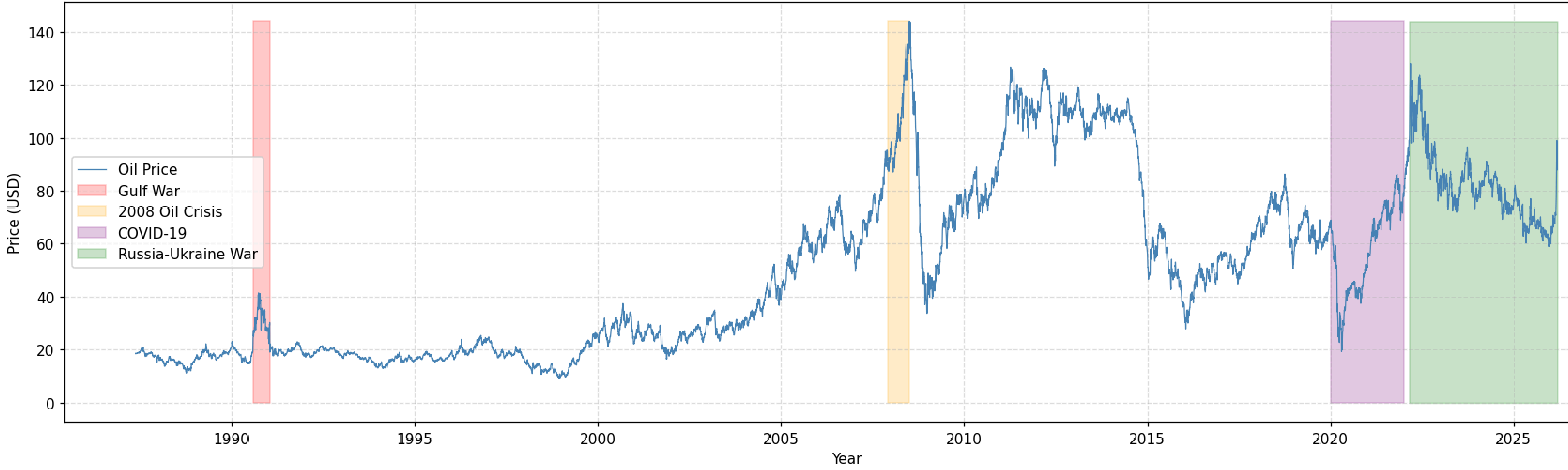
FEATURE ENGINEERING

	Date	Oil Price	Oil Price_lag1	Oil_rollmean7	GPR INDEX_lag1	Return_lag1
0	1987-05-20	18.63	NaN	NaN	NaN	NaN
1	1987-05-21	18.45	18.63	NaN	106.86	NaN
2	1987-05-22	18.55	18.45	NaN	77.52	-0.009709
3	1987-05-25	18.60	18.55	NaN	132.50	0.005405
4	1987-05-26	18.63	18.60	NaN	87.69	0.002692
5	1987-05-27	18.60	18.63	NaN	82.44	0.001612
6	1987-05-28	18.60	18.60	18.580000	134.11	-0.001612
7	1987-05-29	18.58	18.60	18.572857	134.37	0.000000
8	1987-06-01	18.65	18.58	18.601429	141.97	-0.001076
9	1987-06-02	18.68	18.65	18.620000	128.57	0.003760

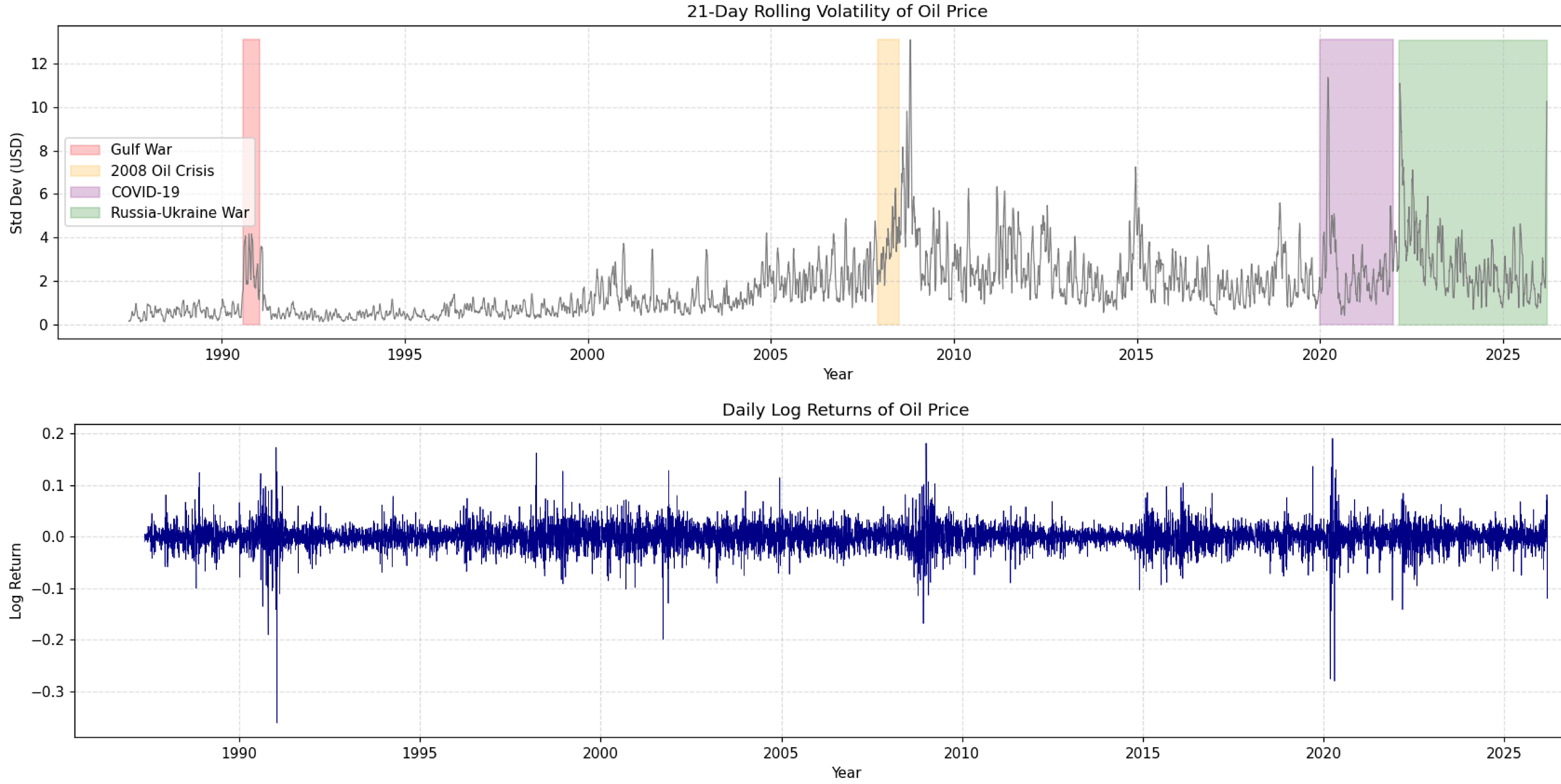
DESCRIPTIVE STATISTICS

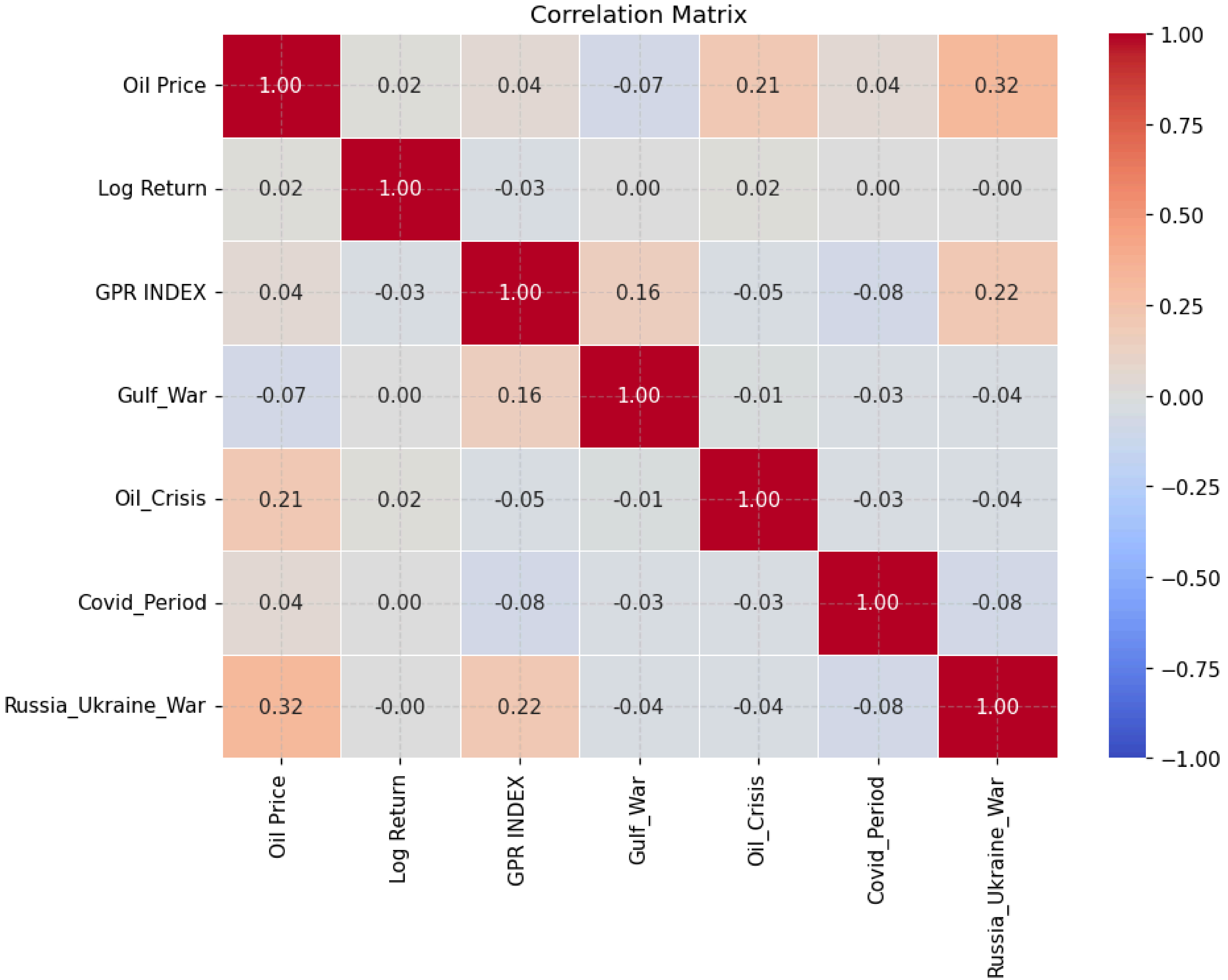


OIL PRICE OVER TIME WITH GEOPOLITICAL EVENTS



ROLLING VOLATILITY & LOG RETURNS TREND

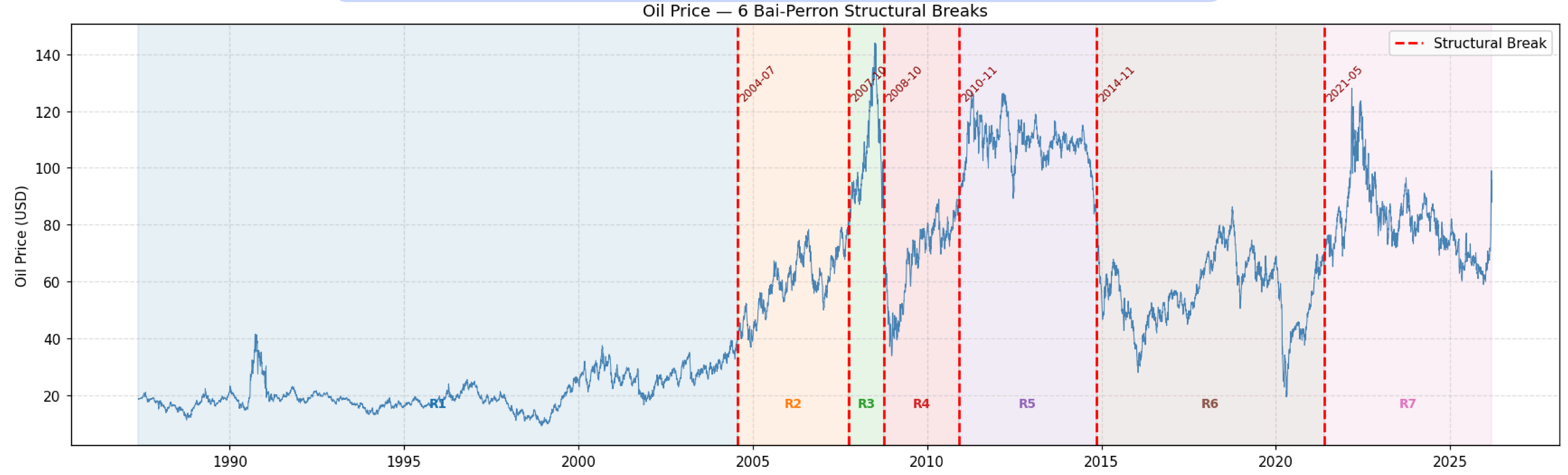


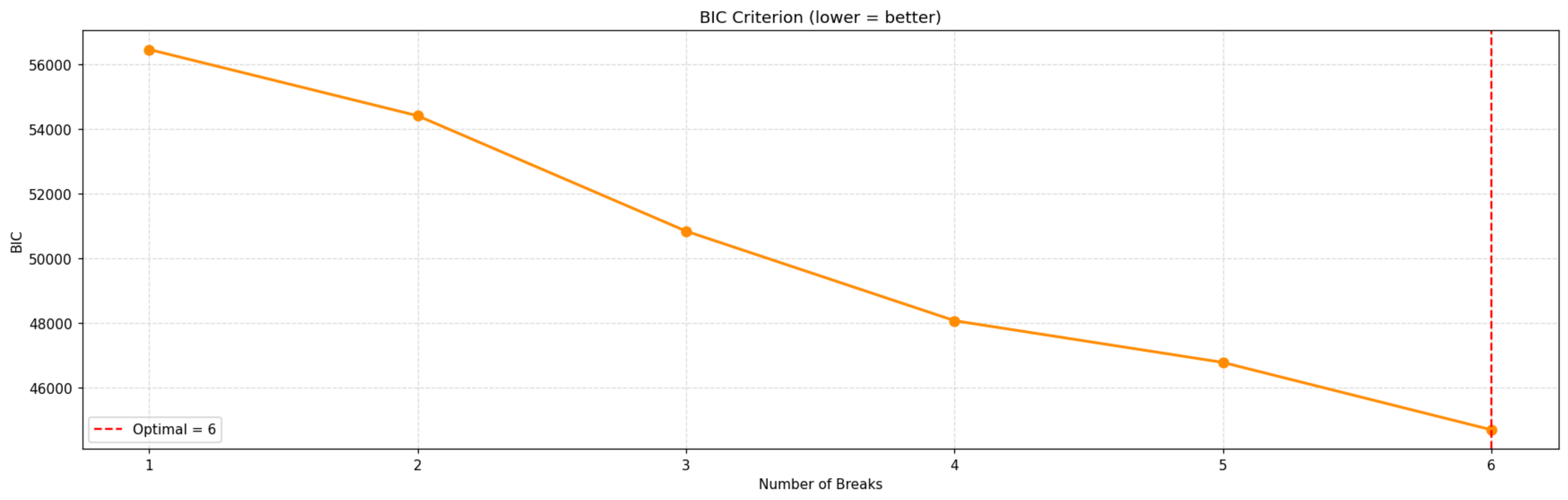

Correlation Matrix
Oil Price
Log Return
GPR INDEX
Gulf_War
Oil_Crisis
Covid_Period
Russia_Ukraine_War
1.00 0.02 0.04 -0.07 0.21 0.04 0.32
0.02 1.00 -0.03 0.00 0.02 0.00 -0.00
0.04 -0.03 1.00 0.16 -0.05 -0.08 0.22
-0.07 0.00 0.16 1.00 -0.01 -0.03 -0.04
0.21 0.02 -0.05 -0.01 1.00 -0.03 -0.04
0.04 0.00 -0.08 -0.03 -0.03 1.00 -0.08
0.32 -0.00 0.22 -0.04 -0.04 -0.08 1.00
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

STATIONARITY TESTS

```
============================================================
  Stationarity Tests: Log Return (Oil Price)
============================================================
           Test  Statistic  p-value  Conclusion
            ADF   -22.6038      0.0 ✅ STATIONARY
Phillips-Perron   -97.0095      0.0 ✅ STATIONARY
           KPSS     0.0342      0.1 ✅ STATIONARY

  ✅ ALL 3 TESTS AGREE: Log Return (Oil Price) is STATIONARY — safe to use in OLS/GARCH.

============================================================
  Stationarity Tests: GPR INDEX
============================================================
           Test  Statistic  p-value      Conclusion
            ADF   -10.0888   0.0000    ✅ STATIONARY
Phillips-Perron   -48.4953   0.0000    ✅ STATIONARY
           KPSS     0.5335   0.0341 ❌ NON-STATIONARY

  ⚠️  MIXED RESULTS for GPR INDEX — consider first-differencing before regression.

  → GPR INDEX may be non-stationary. Created ΔGPR (first difference) for VAR/OLS.

============================================================
  Stationarity Tests: ΔGPR (First Difference of GPR INDEX)
...
Phillips-Perron  -265.9688      0.0 ✅ STATIONARY
           KPSS     0.0069      0.1 ✅ STATIONARY

  ✅ ALL 3 TESTS AGREE: ΔGPR (First Difference of GPR INDEX) is STATIONARY — safe to use in OLS/GARCH.
```

BAI-PERRON MULTIPLE STRUCTURAL BREAK TEST

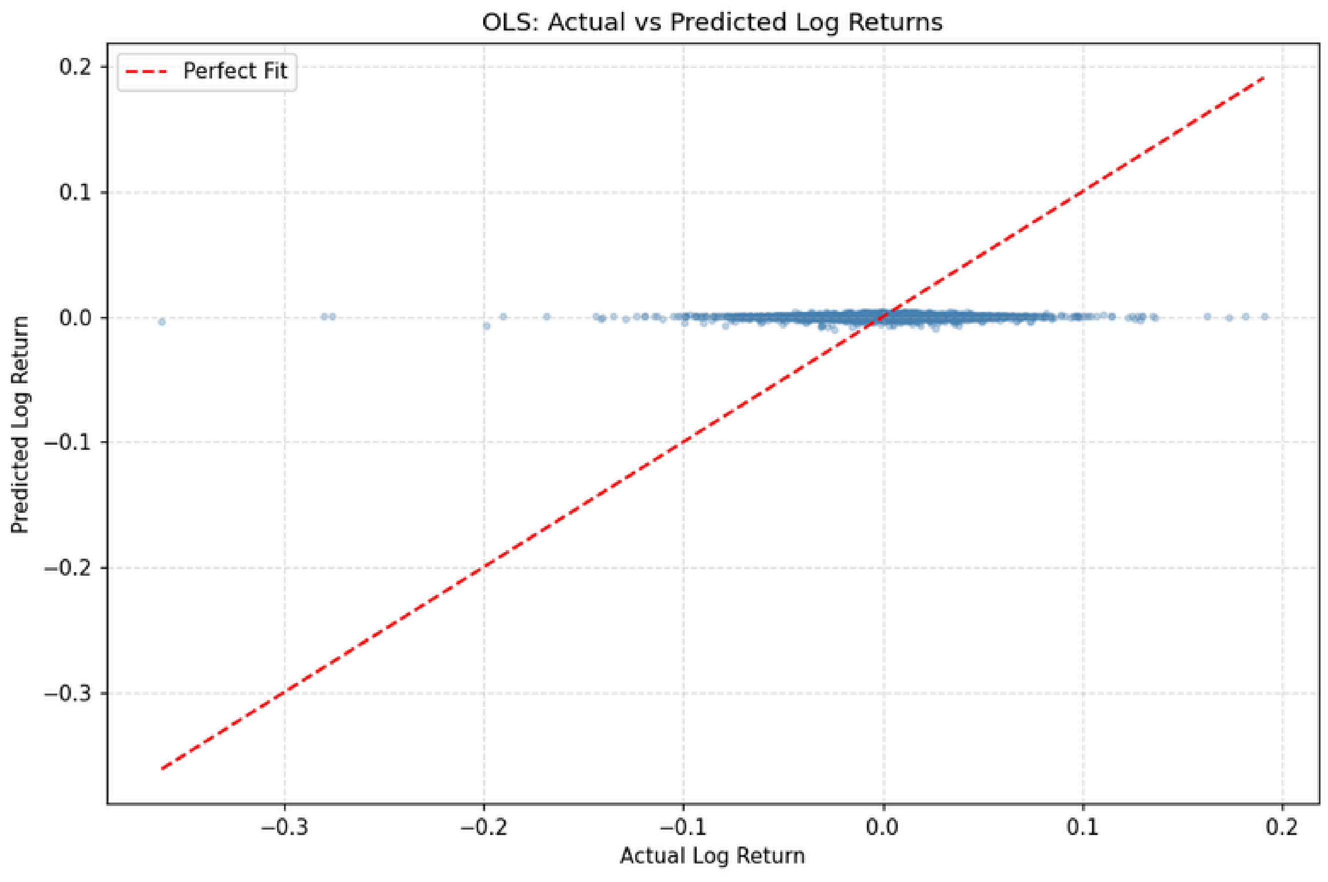




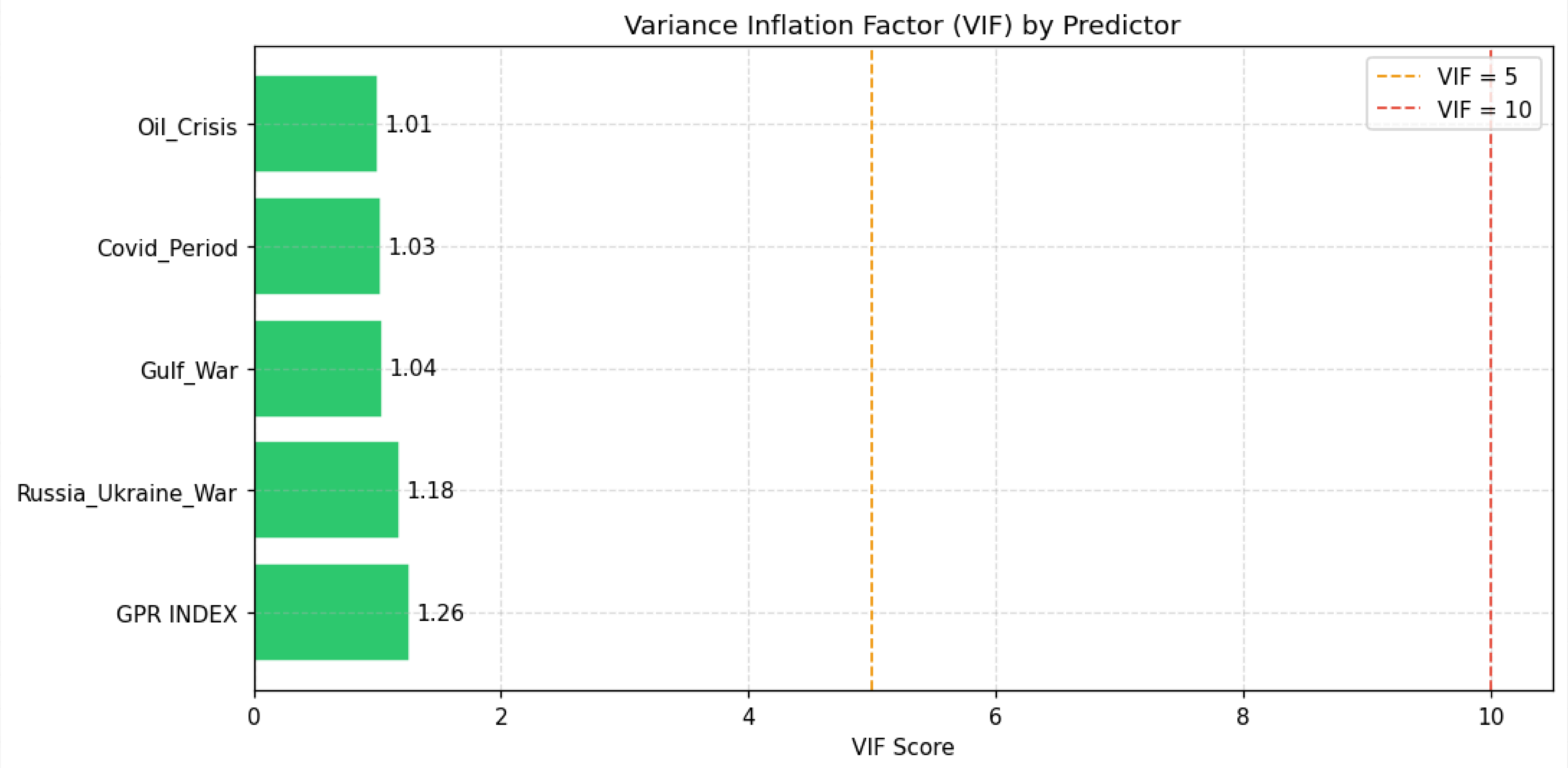

```
BIC Model Selection:
  Breaks                BIC              RSS
----------------------------------------
       1           56453.93       3061382.76
       2           54406.52       2481025.62
       3           50842.49       1723162.65
       4           48077.24       1298131.09
       5           46787.99       1136409.44
       6           44706.76        917812.65

✓  Optimal breaks (BIC): 6
```

OLS REGRESSION

```
                            OLS Regression Results
==============================================================================
Dep. Variable:             Log Return   R-squared:                       0.001
Model:                            OLS   Adj. R-squared:                  0.001
Method:                 Least Squares   F-statistic:                     1.173
Date:                Tue, 07 Apr 2026   Prob (F-statistic):              0.320
Time:                        14:59:31   Log-Likelihood:                 22957.
No. Observations:                9827   AIC:                        -4.590e+04
Df Residuals:                    9821   BIC:                        -4.586e+04
Df Model:                           5
Covariance Type:                  HC3
======================================================================================
                          coef    std err          z      P>|z|      [0.025      0.975]
--------------------------------------------------------------------------------------
const                   0.0013      0.001      1.669      0.095      -0.000       0.003
GPR INDEX           -1.104e-05   7.19e-06     -1.535      0.125   -2.51e-05    3.05e-06
Gulf_War                0.0014      0.006      0.253      0.800      -0.010       0.012
Oil_Crisis              0.0028      0.002      1.673      0.094      -0.000       0.006
Covid_Period         1.202e-05      0.002      0.008      0.994      -0.003       0.003
Russia_Ukraine_War      0.0004      0.001      0.457      0.648      -0.001       0.002
==============================================================================
Omnibus:                     2844.568   Durbin-Watson:                   1.959
Prob(Omnibus):                  0.000   Jarque-Bera (JB):            93170.025
Skew:                          -0.742   Prob(JB):                         0.00
Kurtosis:                      18.011   Cond. No.                     1.19e+03

strong multicollinearity or other numerical problems.
```



MULTICOLLINEARITY TEST (VIF)

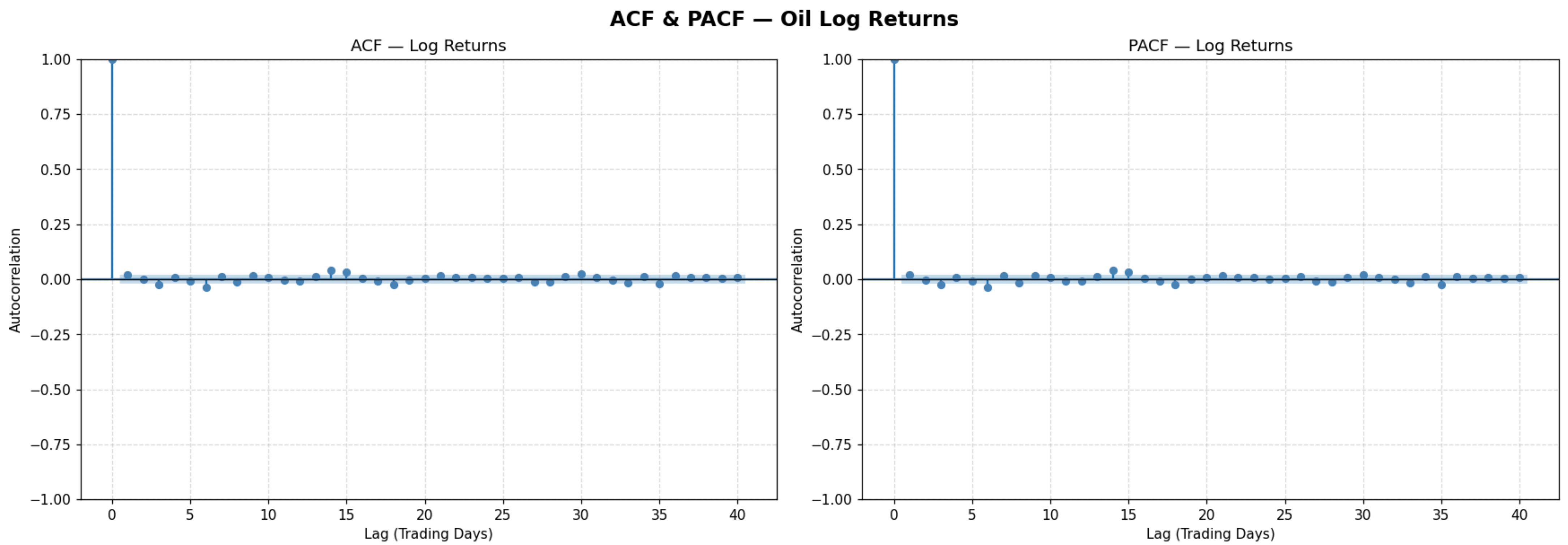


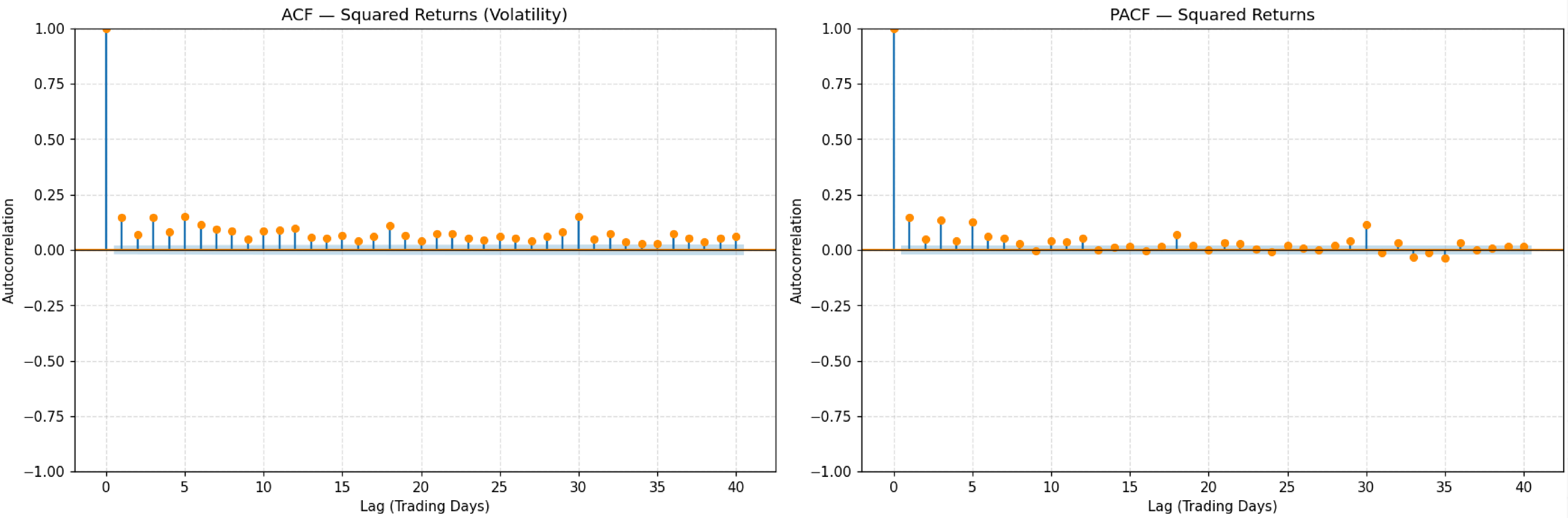
HETEROSKEDASTICITY TEST

```
Heteroskedasticity Tests
==================================================
         Test  LM Stat  p-value Reject H₀?
Breusch-Pagan 369.6883      0.0      Yes ❌
        White 980.2181      0.0      Yes ❌

  → Heteroskedasticity confirmed. HC3 robust SEs used in OLS (Section 9).
    GARCH model (Section 13) directly models time-varying variance.
```

ACF & PACF

LOG RETURNS & SQUARE RETURNS



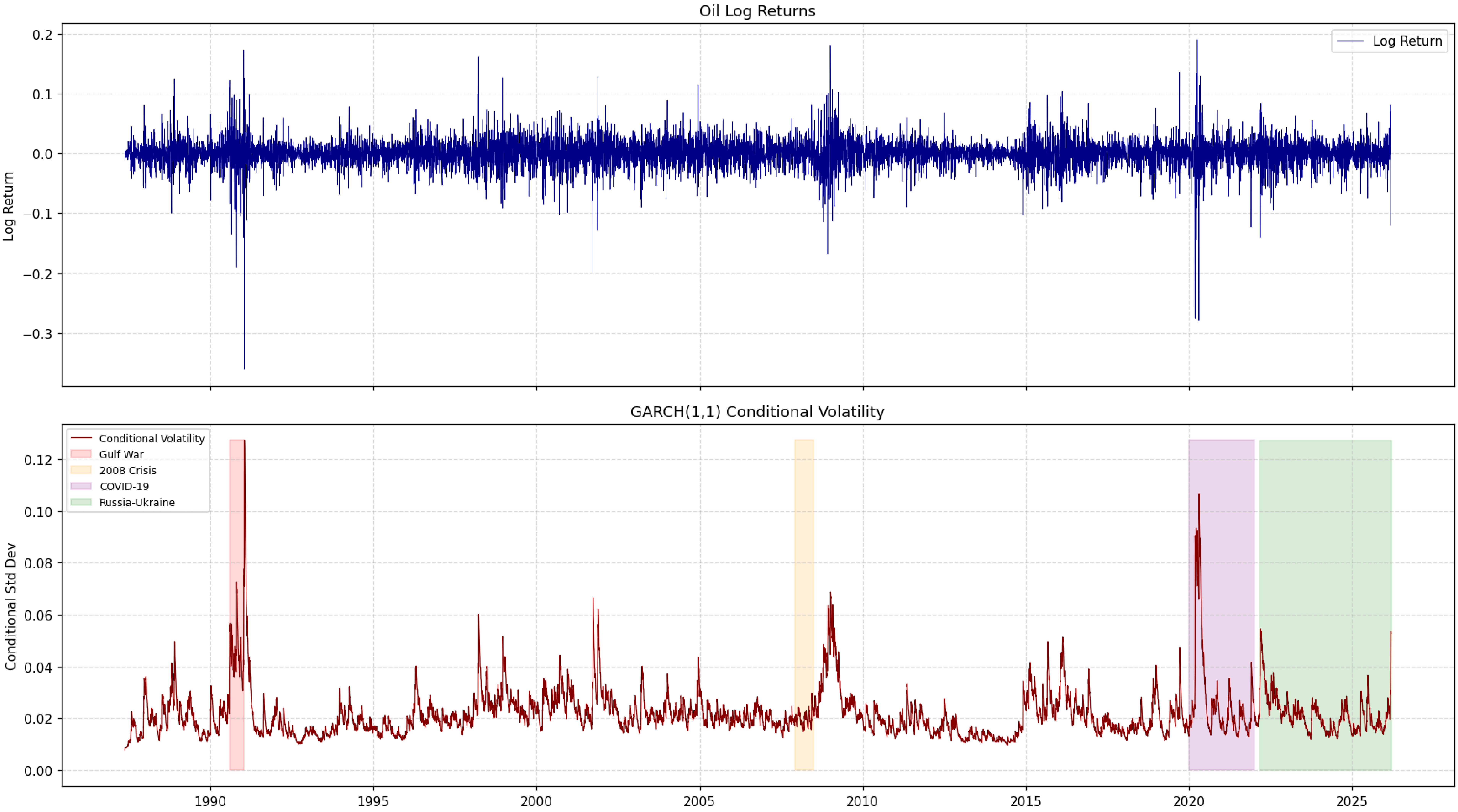



```
95% band: ±0.0198
Significant lags (Returns):        ['Lag 1', 'Lag 3', 'Lag 6', 'Lag 14', 'Lag 15']
Significant lags (Sq. Returns):    ['Lag 1', 'Lag 2', 'Lag 3', 'Lag 4', 'Lag 5', 'Lag 6', 'Lag 7', 'Lag 8', 'Lag 9', 'Lag 10', 'Lag 11', 'Lag 12

• Significant ACF in Sq. Returns → volatility clustering → GARCH justified
```

ARCH & GARCH RESULTS

```
ARCH Test on OLS Residuals
----------------------------------------
ARCH LM Statistic : 702.7232
p-value           : 0.0000

✅ Volatility clustering in residuals confirmed
   GARCH model is strongly justified.
```

```
                         AR-X - GARCH Model Results
==============================================================================
Dep. Variable:                 Log Return   R-squared:                  -0.000
Mean Model:                          AR-X   Adj. R-squared:             -0.001
Vol Model:                          GARCH   Log-Likelihood:           -20897.6
Distribution:                      Normal   AIC:                       41813.3
Method:                Maximum Likelihood   BIC:                       41878.0
                                            No. Observations:             9827
Date:                    Wed, Apr 08 2026   Df Residuals:                 9821
Time:                            02:21:35   Df Model:                        6
                                  Mean Model
======================================================================================
                          coef    std err          t      P>|t|      95.0% Conf. Int.
--------------------------------------------------------------------------------------
Const                   0.0369  6.181e-02      0.598      0.550  [-8.419e-02,  0.158]
GPR INDEX          -3.0845e-05  5.903e-04 -5.226e-02      0.958 [-1.188e-03,1.126e-03]
Gulf_War                0.0316      0.912  3.466e-02      0.972     [ -1.756,  1.819]
Oil_Crisis              0.3160      0.203      1.556      0.120  [-8.195e-02,  0.714]
Covid_Period            0.1778      0.111      1.600      0.110  [-4.000e-02,  0.396]
Russia_Ukraine_War     -0.0815  7.408e-02     -1.100      0.271   [ -0.227,6.373e-02]
                              Volatility Model
============================================================================
                 coef    std err          t      P>|t|      95.0% Conf. Int.
----------------------------------------------------------------------------
omega          0.0680  1.571e-02      4.326  1.521e-05 [3.718e-02,9.878e-02]
...
beta[1]        0.9020  1.133e-02     79.622      0.000     [  0.880,  0.924]
============================================================================

Covariance estimator: robust
```



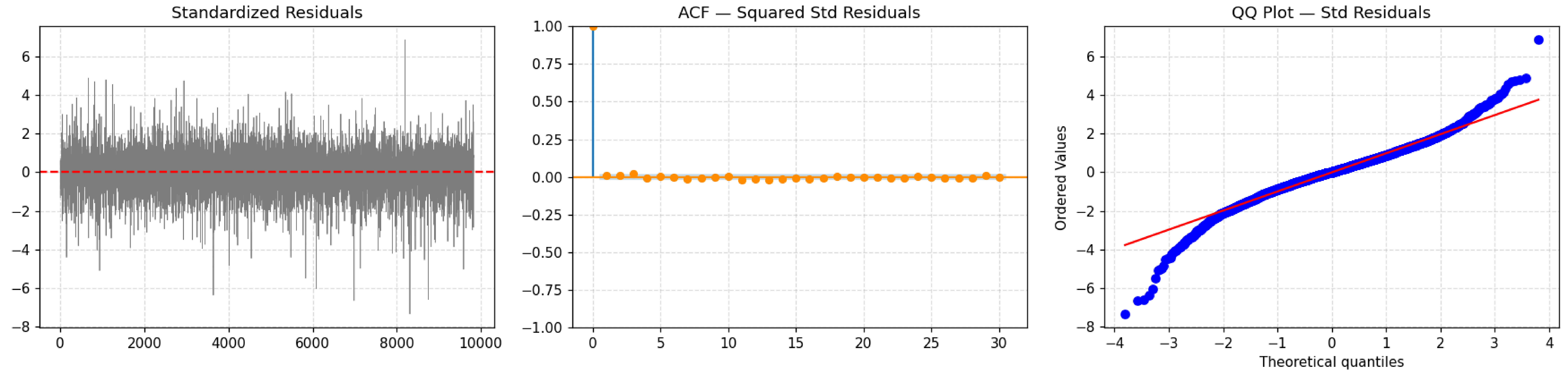
Oil Log Returns
Log Return
Log Return
GARCH(1,1) Conditional Volatility
Conditional Volatility
Gulf War
2008 Crisis
COVID-19
Russia-Ukraine
Conditional Std Dev

GARCH VS GJR-GARCH VS EGARCH COMPARISON

```
Model Comparison by Information Criteria
====================================================
           Model       AIC       BIC  Best AIC  Best BIC
      GARCH(1,1) 41813.25 41877.99     False     False
GJR-GARCH(1,1,1) 41796.66 41868.59      True     False
     EGARCH(1,1) 41835.38 41864.16     False      True

✅  Best model by AIC: GJR-GARCH(1,1,1)

GJR-GARCH Summary (asymmetry term γ):
                        AR-X - GJR-GARCH Model Results
==============================================================================
Dep. Variable:              Log Return   R-squared:                      -0.000
Mean Model:                       AR-X   Adj. R-squared:                 -0.001
Vol Model:                   GJR-GARCH   Log-Likelihood:               -20888.3
Distribution:                   Normal   AIC:                           41796.7
Method:             Maximum Likelihood   BIC:                           41868.6
                                         No. Observations:                 9827
Date:                 Wed, Apr 08 2026   Df Residuals:                     9821
Time:                         02:21:36   Df Model:                            6
                                      Mean Model
=====================================================================================
                         coef    std err          t      P>|t|       95.0% Conf. Int.
-------------------------------------------------------------------------------------
Const                  0.0161  6.505e-02      0.248      0.804     [ -0.111,  0.144]
...
beta[1]        0.9028  1.117e-02     80.853      0.000     [  0.881,  0.925]
============================================================================
```

GARCH(1,1) RESIDUAL DIAGNOSTICS

```
Ljung-Box Test on GARCH Standardized Residuals
H₀: No serial correlation (p > 0.05 is GOOD)
=======================================================

On residuals (mean equation adequacy):
    lb_stat  lb_pvalue
5   13.7866     0.0170
10  16.5542     0.0848
20  28.7945     0.0919

On squared residuals (variance equation adequacy):
    lb_stat  lb_pvalue
5    6.8487     0.2321
10  10.0471     0.4364
20  21.6840     0.3579
```



VAR MODEL + IRF

```
VAR dataset: 9827 observations
Date range: 1987-05-21 → 2026-03-12

VAR Lag Order Selection (IC):
 VAR Order Selection (* highlights the minimums)
==================================================
       AIC         BIC         FPE         HQIC
--------------------------------------------------
0      -0.1594     -0.1579      0.8527     -0.1589
1      -0.3423     -0.3379      0.7101     -0.3408
2      -0.3909     -0.3836      0.6765     -0.3884
3      -0.4151     -0.4049      0.6603     -0.4116
4      -0.4347    -0.4215*      0.6475     -0.4302
5      -0.4362     -0.4201      0.6465     -0.4308
6      -0.4378     -0.4188      0.6454    -0.4314*
7      -0.4383     -0.4163      0.6452     -0.4308
8      -0.4391     -0.4142      0.6446     -0.4307
9     -0.4407*     -0.4129     0.6436*     -0.4313
10     -0.4404     -0.4097      0.6438     -0.4300
--------------------------------------------------

✅ BIC-selected lag order: 4
  Summary of Regression Results
==================================
Model:                         VAR
...
ΔGPR             -0.006403  1.000000
```